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Magnetostratigraphy and ^{40}Ar–^{39}Ar geochronology of the Malwa Plateau region (Northern Deccan Traps), central western India: Significance and correlation with the main Deccan Large Igneous Province sequences

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ABSTRACT

Basaltic lava flows of Malwa Plateau together with the Mandla area and the Toranmal section constitute the Northern Deccan Province in central western India. With the present outcrop area of 80,000 km^2 the Malwa flows form an extensive but relatively thin sequence.

We sampled Deccan basalts along two 100 km each N–S and E–W transects along the western and southern rims of the Malwa Plateau. The 67.12 ± 0.44 Ma ^{40}Ar–^{39}Ar age obtained in the present study clearly shows that the eruption of the Malwa flows started during the magnetochron C30n. By mapping the magnetic C30n/C29r transition on the western and southern edge of the plateau, the total volume and eruption rate can be calculated (42,000 km^3; 0.03 km^3/a). Furthermore the results point towards the Narmada-Tapi rift zone as the possible source area for the Malwa lava flows.

The Malwa sequence is older and partly synchronous with the thick sequences of the Western Ghats. Since their eruption centers are spatially independent, they should not be correlated with younger, although geochemically similar, formations of the Western Ghats. Other Northern Deccan flows show clear similarities with the Malwa flows and can be magnetostratigraphically and geochemically correlated. We therefore suggest a new subgroup for the Northern Deccan Province, the "Narmada Subgroup", defined on the geochemical and magnetic characteristics of the Malwa, Mandla and Toranmal flows. The introduction of this new subgroup clarifies the position of the Northern Deccan Province in the Deccan stratigraphy and is a step towards a better understanding of the Deccan eruption history.

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1. Introduction

Continental flood basalts (CFB) and corresponding submarine oceanic plateaus (OP) have received much attention during the past decades, as majority of mass extinctions and oceanic anoxic events can be linked to the eruption of huge volumes of lava in Large Igneous Provinces (e.g. Vogt, 1972; Rampino and Stothers, 1988; Courtillot, 1994; Wignall, 2001; Courtillot and Renne, 2003). The Cretaceous–Paleogene age Deccan Large Igneous Province (LIP) in central western India is crucial in this context as timing of eruption of lava corresponds to the climate crisis and consequent mass extinction (e.g. McLean, 1981; Officer and Drake, 1983; Officer et al., 1987; Keller et al., 2009a; Chalapathi Rao et al., 2014). The Mid Phanerozoic volcanic eruptions have been associated with mass extinction events, namely the end-Guadalupian extinction–Emeishan flood basalts, the end-Permian extinction–Siberian Traps, the end-Triassic extinction–central Atlantic volcanism and the early Toarcian extinction–Karoo Traps (e.g. Wignall, 2001; Zhou et al., 2002; Benton and Twitchett, 2003; Jourdan et al., 2008; Schoene et al., 2010).

The timing and duration of the volcanism are crucial factors for the impact on the atmosphere. Several attempts were made to obtain a precise age and time frame for the Deccan LIP. A concept of a pin-pointed date for the Deccan LIP as a single event was soon discarded, as LIPs are characterized by multiple eruptive pulses (Keller et al., 2009b and references therein).

Earlier ^{40}K–^{40}Ar (e.g. Wellman and McElhinny, 1970; Kaneoka and Haramura, 1973) and later ^{40}Ar–^{39}Ar ages (e.g. Duncan and Pyle, 1988; Courtillot et al., 1986; Hofmann et al., 2000; Chenet et al., 2007; Hooper et al., 2010; Chalapathi Rao et al., 2011) cluster at 66 ± 2 Ma. There is magnetostratigraphic evidence that a major

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part of the 2000 m thick Western Ghats (e.g. Vandamme and Courtillot, 1992 and references therein) lavas erupted within the reversed C29r chron (700,000 years duration: Gradstein et al., 2012).

After the breakup of Gondwana Greater India experienced two major rifting events with associated volcanism leading to the early Cretaceous (~88 Ma Madagascan volcanism, Storey et al., 1995; 83.6–91.4 Ma, St. Mary's Island, Torsvik et al., 2000) separation of Madagascar and the end Cretaceous (~65–64 Ma: Hooper, 1990; Seychelles volcanism at 61.3–66.8 Ma: Ganerød et al., 2011) separation of the Seychelles microcontinent. The latter is coeval with the Deccan volcanism.

Most of the CFBs have conventionally been related to plume volcanism (Morgan, 1972, 1983; Richards et al., 1989; White and McKenzie, 1989; Campbell and Griffiths, 1990). These short-duration, large volume igneous events occur when the hot and buoyant plume head interacts with the lithosphere. The continuing magmatism associated with the remaining plume conduit or "tail" is than represented by a hot spot. This model relates the main part of the Deccan LIP to a massive "plume head" and the Lakshadweep–Chagos Ridge and the volcanic chain of Maldives as the trace of the plume tail. The presently volcanically active Réunion island in the Indian Ocean would then represent the current position of the plume (Duncan and Pyle, 1988). This model has been questioned by Sheth (e.g. 2005, 2007) who suggests a "passive", non-plume related model for the Deccan volcanism. In his model he proposed extensive decompression melting, when the Central Indian Ridge jumped onto the northerly drifting Greater India, splitting the Seychelles. An alternative non-plume related model for CFBs was also introduced earlier by King and Anderson (1995). They associated the locations of flood basalts with boundaries of thick and thin lithosphere. According to them those non-uniform boundary conditions are able to induce a shallow sub horizontal flow of hot material from under the thicker boundary layer towards the non-uniform boundary. An externally imposed breakup of the lithospheric plates can than trigger the effusion of the flood basalts. In yet another model, Chatterjee (1997) presumed that the impact of an asteroid in the Indian Ocean (Hartnady, 1986), coincident with the Cretaceous/Paleogene transition, could be the reason for the breakup of Greater India that also triggered the eruption of the flood basalts.

As it is usual for LIPs, the Deccan Flood Basalts can be divided into spatial domains. The Deccan Provinces and the most prominent rift zones associated with the Deccan LIP are shown in Fig. 1.

Deccan Traps have been extensively studied during the last decades. However, most of these studies focus on the Western Ghats, especially the Pune–Mahabaleshwar section where a ~2000 m thick sequence of lava flows is exposed. The Northern Deccan Province, comprising the Mandla, Malwa, Toranmal and Pavagadh sections have not received much attention despite covering a significant geographic terrain. The chronological and genetic relationship of the voluminous southern province with the northern segments need to be evaluated and correlated, being crucial for the understanding of the tectonomagmatic history of the Deccan LIP. This paper provides new ^{40}Ar–^{39}Ar geochronologic and magnetostratigraphic data from the Malwa Plateau (Fig. 1). The results are discussed in terms of age and duration of volcanism and correlation with the Deccan Traps from the type area, the Western Ghats.

2. Geological overview

The Western Ghats sequences have been used to work out the stratigraphy of Deccan Traps mainly on the basis of geochemical characteristics of the (mostly thoeliitic) lava flows (Cox and Hawkesworth, 1985; Devey and Lightfoot, 1986). The geochemical characteristics used to distinguish the main Deccan formations are compiled in Table 1. The geochemical differences of the major element compositions are controlled mainly by the degree of gabbro fractionation, while crustal contamination can have a strong effect on trace elements and Sr-isotopes (Cox and Hawkesworth, 1985).

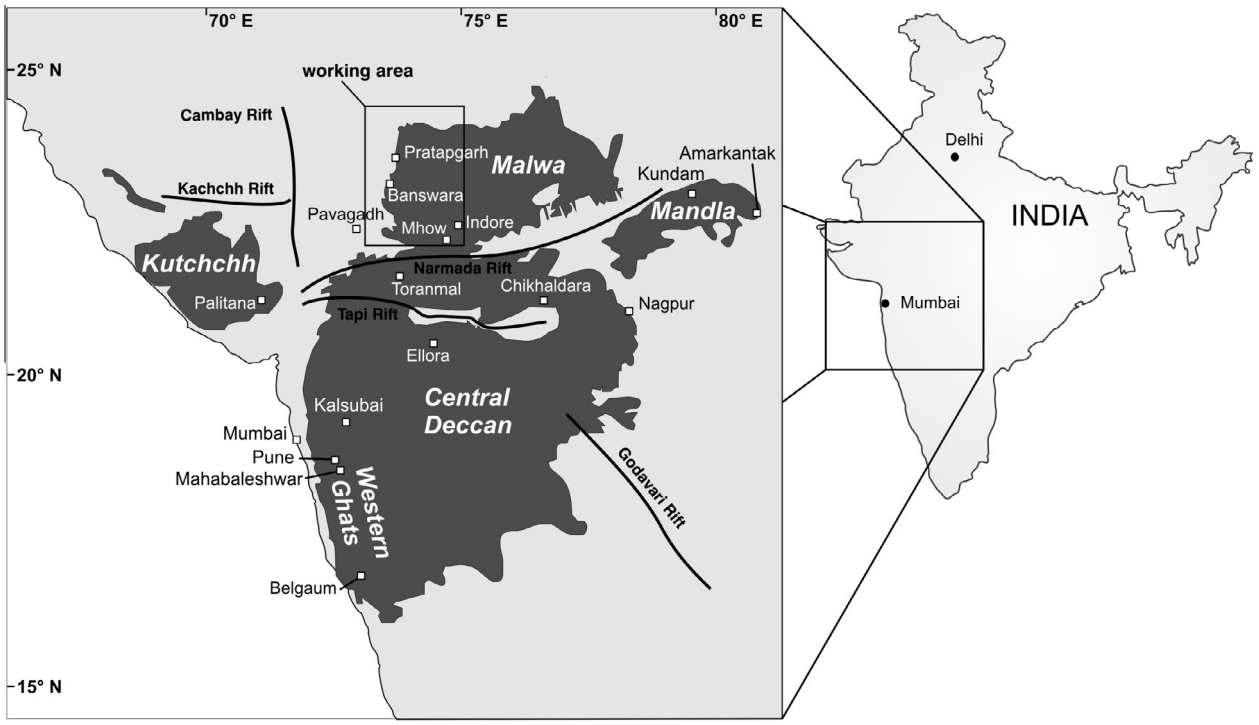


Fig. 1. Map showing extent of Deccan LIP in central western India. Sites relevant for this study and prominent rift zones of the Indian subcontinent, which might be correlated with the Deccan LIP are also shown.

Table 1
Relevant geochemical characteristics of main formations of the Southwestern Deccan LIP. Data from Devey and Lightfoot (1986), Lightfoot et al. (1990) and Jay and Widdowson (2008).

Formation	Sr (ppm)	Ba (ppm)	Ba/Y	$^{87}Sr/^{86}Sr$	Zr/Nb
Panhala	<200	<90	–	–	>13
Mahabaleshwar	>250	>100	>4	>0.705	<10.5
Ambenali	200–250	<100	<3.5	<0.705	10.5–15
Poladpur	<240	>100	>3.5	0.705–0.712	15–20
Bushe	–	>100	–	>0.713	>20

Naturally, there are constraints to correlate the northern Deccan districts with the southern geochemical formations (e.g. Sreenivasa Rao et al., 1985; Peng et al., 1998; Mahoney et al., 2000; Khadri, 2003). A summary of geochemical and magnetostratigraphic studies is shown in Fig. 2. A trivial linkage of the northern and southern lava flows seems to be missing. Although there are similarities in the geochemical fingerprints between Northern and Southern Formations, a clear correlation is not tenable. Lava flows from the Mandla lobe, Chikaldara and Mhow sections show isotopic and geochemical similarities with the Ambenali, Poladpur and Khandala Formations, however, they possess significantly higher $^{206}Pb/^{204}Pb$ ratios as compared to the southwestern flows (Peng et al., 1998). Therefore it can be considered that they may have erupted from different feeder system(s). The Toranmal section in the Narmada region exhibits flows similar to the Poladpur Formation of the Western Ghats (e.g. Mahoney et al., 2000). However, a clear correlation with the main Deccan stratigraphy was not possible for the authors. Also the thick pile of Deccan lavas at the Mount Pavagadh cannot be decidedly correlated to the Southern Deccan Stratigraphy. Sheth and Melluso (2008) mentioned that the normally magnetized Pavagadh sequence is probably older than the Jawhar Formation (oldest formation in the Western Ghats stratigraphy). The lava sequence of the Kachchh area in the western part of the Deccan LIP shows a similar petrogenetic evolution rather than stratigraphic or genetic relationships (Sheth et al., 2013).

There is evidence from paleomagnetic and radiometric ages that the basal lava flows of the northern area (Narmada region) erupted earlier than the major part of the Western Ghats series. Moreover some of the northeastern profiles show a transition from a normal magnetochron to a reversed magnetochron while this transition usually does not occur in southern sequences. In the Western Ghats the transition from the reversed magnetochron C29r to the younger normal magnetochron C29n is common and can be observed in the Mahabaleshwar Formation (e.g. Chenet et al., 2008 and references therein). Due to the differences in geochemical composition of various Deccan districts, Sheth et al. (2013) concluded that major chemical magma types of the Deccans are not chronostratigraphic, instead erupted at different times in different regions of the LIP. This is an important finding, as many recent studies still include the Northern Deccan Province in the Western Ghats geochemical stratigraphy.

3. Sampling strategy and methods

3.1. Sampling area

The Malwa Plateau forms the northern part of the Deccan Traps. The sampling area and the sampling sites are shown in Fig. 3. With an area of about 80,000 km^2 (whole Deccan: ~500,000 km^2;

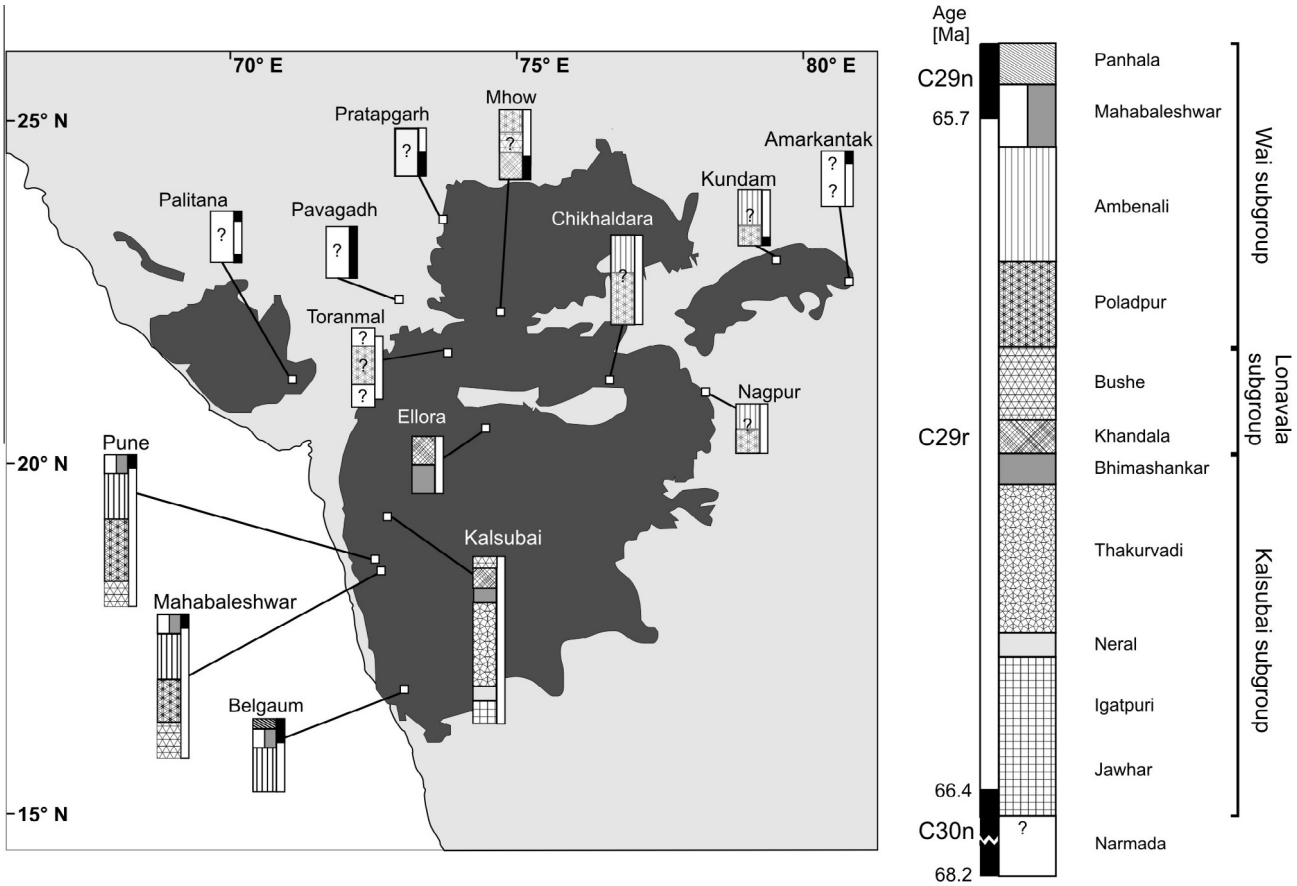


Fig. 2. Map showing Deccan stratigraphy (compiled from Vanderkluysen et al., 2011; Peng et al., 1998; Sheth and Melluso, 2008; Sheth et al., 2013); magnetic polarity after Pande (2002) and references therein, Paul et al. (2008), Courtillot et al. (1986), Verma and Pullaiah (1971) and this study. Magnetic Time scale after Gradstein et al. (2012); K/T boundary age after Kuiper et al. (2008).

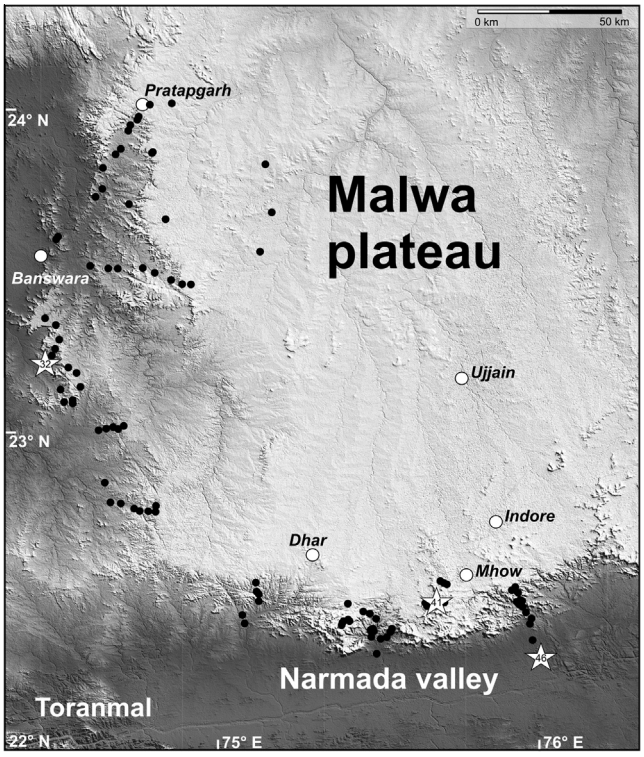


Fig. 3. Map showing sampled sites (filled circles). The asterisks mark the sites for samples used for ^{40}Ar–^{39}Ar dating.

Mahoney, 1988; Chalapathi Rao et al., 2011) and an average elevation of 500 m above mean sea level (a.m.s.l.) the plateau represents an important but until now almost unaccounted part of the Deccan Flood Basalt Province in India. The topography in vast areas of the plateau is flat, only the brim in the western and southern parts of the Malwa region shows some slope, giving an insight into the architecture of different lava flows building up the landscape and allowing sampling over the whole profile. In the very south, close to Indore city, some hills (e.g. Singachori Hill) on the plateau define the highest points of the Malwa (860 m a.m.s.l.). The contact to the underlying basement (mostly from the Paleo Meso Proterozoic Bijawar Group and the Paleo-Mesoproterozoic Vindhyan Supergroup) occurs at different altitudes. While lava flows in the south can be found in the Narmada valley (going down to about 200 m a.m.s.l.) the contact is not always visible due to the underlying basement. On the western brim, the contact to the basement varies in altitude between 280 and 330 m a.m.s.l. This gives an average thickness of Deccan basalts of ~500 m in the southern and ~200 m in the western parts of the plateau.

The Malwa Plateau received little attention so far. Paleomagnetic studies were carried out by Bhalla and Rao (1974), Rao and Bhalla (1981), Khadri and Nagar (1994) and Khadri (2003) while geochemical characteristics were described by Sreenivasa Rao et al. (1985) and Peng et al. (1998). These studies were carried out in the southern part of the Deccan Volcanic Province. No details are available for the western part of Malwa Plateau except some preliminary geochemistry and flow delineation studies by Sinha-Roy and Shukla (1988), Srivastava et al. (1988), Pandit and Srivastava (1998).

3.2. Lava flow characteristics

Two of the most important factors influencing the rheology of lava flows are the effusion rate and the viscosity. The volcanics of LIPs are characterized by high effusion rates and low viscosity, therefore, individual lava flows cover long distances (e.g. Ferrar LIP: Elliot et al., 1999; Deccan: Self et al., 2008).

Basaltic lavas extruding at relatively low effusion rates usually form thick flows, consisting of a number of small flow units known as pahoehoe lobes. These lobes overrun each other as anastomosing channels and can amalgamate by cooling. Pahoehoe flows are technically always compound flows and a separation of the individual flow units is often not possible. The time interval between successive flow units is normally short (minutes to months) so no vegetation can settle on the surface of the flow unit. The Hawaii type flows show the characteristic pahoehoe morphology like hummocky shape, inflated sheets and tumili (pressure ridges) and are called 'compound' flows (Walker, 1971). When the effusion rate is higher, the lava produces extensive flows and forms large-volume flow sheets. They congeal as single cooling unit and cannot be differentiated into individual flow units. Such lava flows are called 'simple' flows. In most flood basalt provinces both 'compound' and 'simple' lava flows are present, but typically the most extensive and far-reaching flows are the simple flows, formed during high extrusion rate (Walker, 1971). This basic classification of Walker (1969) into 'simple' and 'compound' flows is still relevant and applicable to the Deccan Traps. While compound flows are more common in the Western Ghats, Pune and Kachchh sections, simple lava flows are predominantly seen in the south eastern parts, the Toranmal section and Mahabaleshwar (Bondre et al., 2004). The Malwa Plateau also consists of mainly simple lava flows, where the thickness of individual flow typically varies between ten and twenty meters. Due to the low relief, poor exposures and partially strong vegetation at most parts of the Malwa, it is very difficult to trace the flows over a longer distance, but it seems that, the thickness of the flows remains more or less constant over a respectable distances. An ideal simple lava flow displays three zones; a crustal, central and a basal zone. The crustal zone is usually the most weathered part of the flow, however, in some cases this zone is not easy to recognize, because the lava flows are amalgamated and the crust is not preserved. This happens when the time interval between the lava flows is short. If the time gap is longer, the crustal zone can reach up to several meters. Red horizons (Fig. 4: red boles) are characteristic of the top zone of the flow and can represent sediments, weathered basalts, weathered pyroclasts, initial oxisols or metasomatized/baked zones (for references see Chenet et al., 2007). The red boles can show various characteristics but are often traversed by honeycomb gypsum.

Progressing to the center of the flow, the weathering and the reddened oxides gradually decrease. Some flows show a flow top breccia (Fig. 4) which can reach a few meters and differs clearly from the other parts of the flow. The flow top breccias contain highly vesicular and amygdaloidal fragments in a groundmass of fine and often glassy material. The crustal zone has a very complex and uneven cooling history. Columnar jointing is a rare phenomenon even if a crustal zone is present. Although the crustal zone has a completely different rheological and cooling history than the rest of the flow, the transition to the central zone is often hard to spot in the field. The central part of a flow is typically the thickest part of a single flow. It can easily reach tens of meters. The habitus is often dominated by columnar jointing, spheroidal weathering and massive zones. A three-tiered (lower colonnade, middle entablature, upper colonnade) structure of central zone is seldom established. Only thick flows show different zones in their central domain. Most flows are dominated either by a colonnade or an entablature zone. In the colonnade, the columns are usually well developed, with a polygonal (ideally hexagonal) cross section. The columns can have a diameter of 0.1 m to over 1 m and can reach a length of more than 4 m. Multi-tiered columns are common, and in many cases they are separated by platy joints

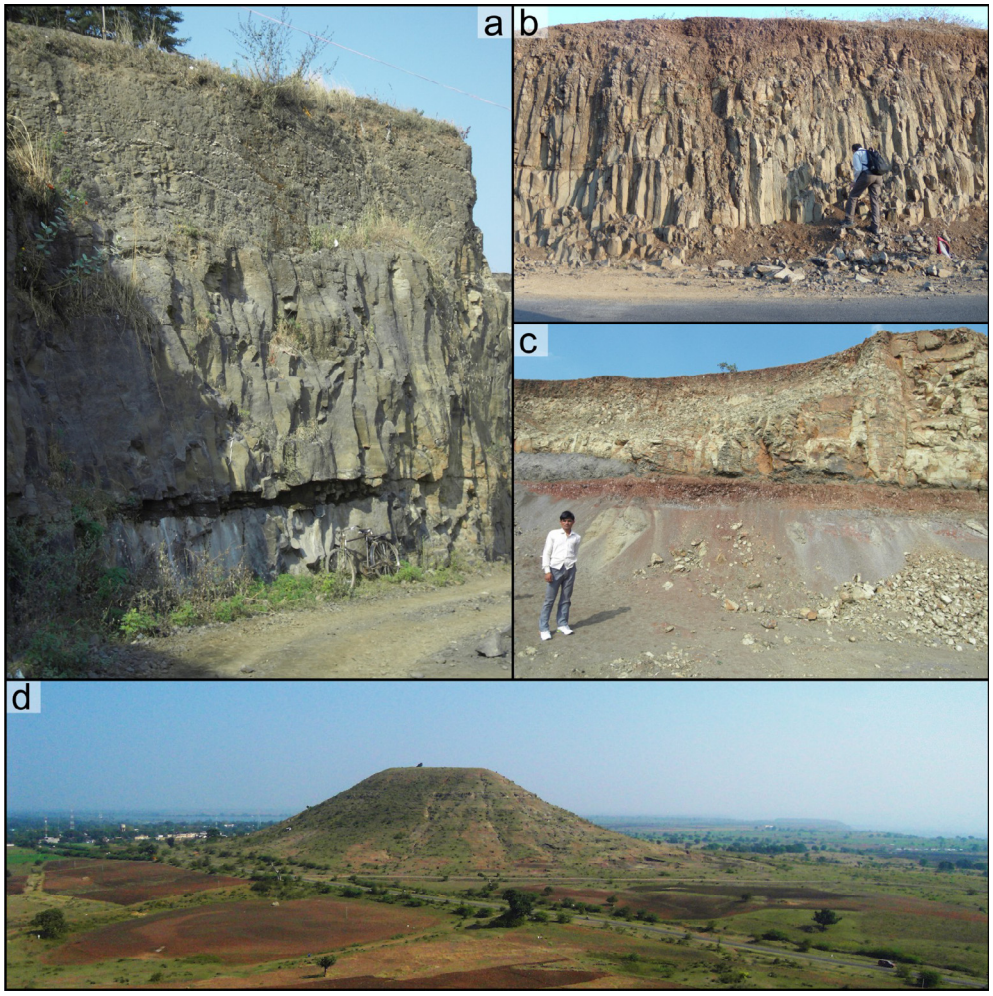


Fig. 4. Typical outcrops of the Malwa flows. (a) Simple flow with well-developed flow top breccia and multi-tiered columns separated by platy joints. Find bicycle as scale. (b) Simple flow with well-developed multi-tiered columns. (c) Red bole horizon overlain by a weathered simple flow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Fig. 4). The lava flows lie more or less horizontal. The tilt is in the most areas is less than 1°. In the western part of the plateau some normal faults are present, which can be identified by mapping the elevation of the magnetic reversals (magnetostratigraphic mapping). Although the habitus of the single flows is quite homogenous, the grain size can vary from a very fine grained to an almost megacrystic fabric. The phenocrysts are mainly plagioclase and clinopyroxene, minor orthopyroxene and iddingsite (altered olivine). All lava flows contain a significant fraction of titanomagnetite. The titanomagnetites crystalize in various shapes (Fig. 5) and resemble the main magnetic carrier of the tholeiitic lava flows. Like most parts of the traps, the Malwa lavas suffer from weathering. Although some areas are quite weathered, it was possible to collect suitable samples from most of the sites. BSE (backscattered electron) images of representative samples can be seen in Fig. 5.

Samples, mainly from the central parts of flows were collected along the western and southern slopes of Malwa Plateau (Fig. 3). Oriented hand samples were collected primarily for magnetic fabric analysis. The samples were used for a simple magnetic stratigraphy. Reversals of the Earth's magnetic field can be used as an isochronal marker horizon on a supraregional scale. After preliminary studies sampling was intensified at an altitude of 300–450 m to map the magnetic transition. Fig. 6 shows the sampled sites along the N–S and W–E profiles in respect to the altitude and the longitude/latitude, respectively. For the radiometric analysis only visibly "fresh" samples with large feldspar phenocrysts were chosen. The samples for radiometric dating come from the very bottom (site 32 in the west; site 46 in the south) and the very top (site 41 on top of Singachori Hill) of the sequence (Fig. 6).

3.3. ^{40}Ar–^{39}Ar age determination

Samples from three sites (s32, s41 and s46) were chosen for radiometric ^{40}Ar–^{39}Ar age determination. All samples are tholeiitic basalts characterized by large feldspar phenocrysts. Samples were crushed with a jaw crusher and sieved (mesh size 180–250 μm).

Conventional techniques were employed to separate feldspars. The mineral separates and whole rock samples were washed in distilled water and acetone several times and finally handpicked under a binocular microscope. Mineral grains with grain coatings or inclusions were avoided. The samples were packed in aluminum capsules together with the Taylor Creek Rhyolite (TCR) flux monitor standard along with pure (zero age) K_2SO_4 and CaF_2 salts. The samples were irradiated together at IFE (Institute for Energiteknikk, Kjeller, Norway) for 72 h with a nominal neutron flux of 1.3×10^{13} n (cm^{-2} s^{-1}). The correction factors for the production of isotopes from Ca were determined to be $(^{39}Ar/^{37}Ar)_{Ca} = (5.9492 \pm 0.1531) * 10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = (2.3339 \pm 0.1045) * 10^{-4}$ and $(^{40}Ar/^{39}Ar)_{K} = (1.586452 \pm 0.0178291) * 10^{-1}$ for the production of K (errors quoted at 1σ). The samples were step heated using

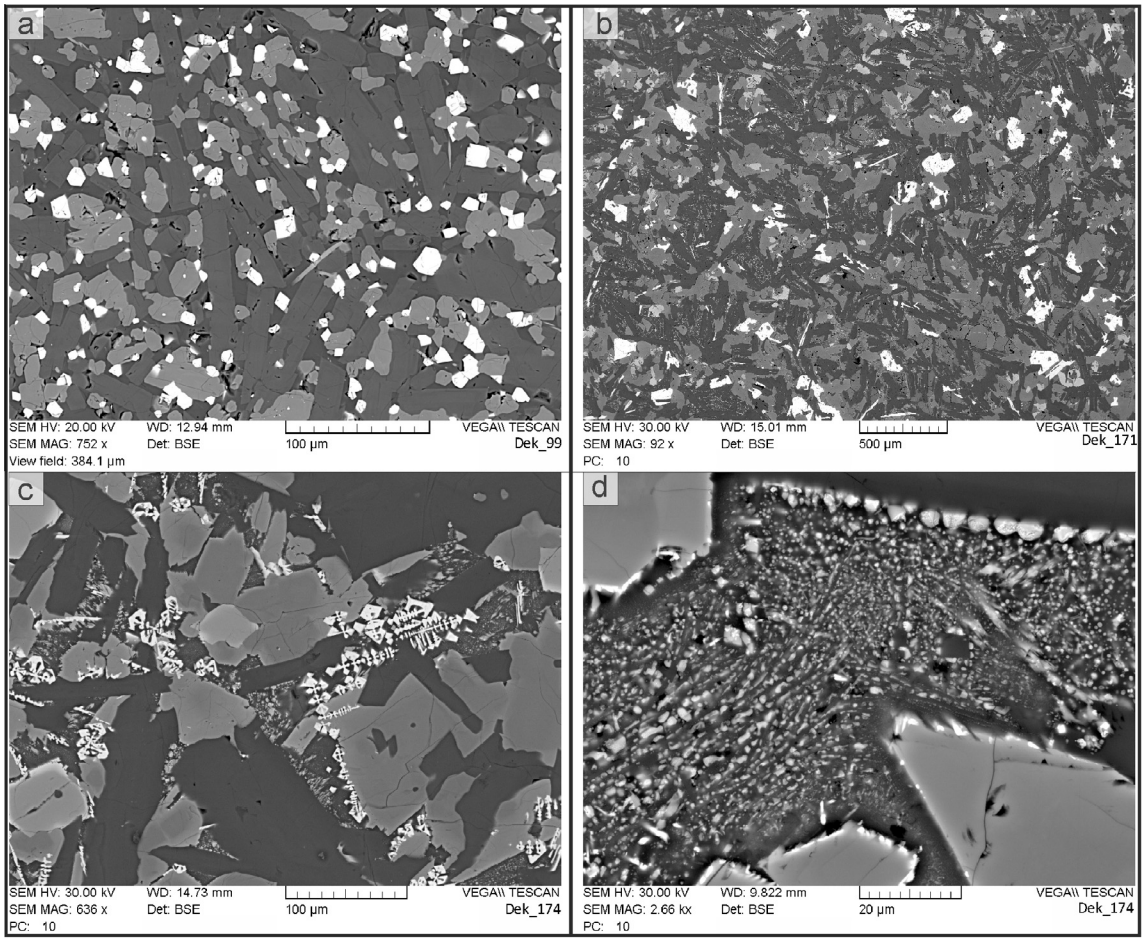


Fig. 5. BSE images of typical Malwa lava flows. Especially tmt crystallizes in various shapes: (a) euhedral, (b) subhedral, (c) skeletal, (d) or as tiny "drops".

a defocused Merchantek CO_2 laser. The extracted gases were passed over SAES GP-50 getters for the first 2 min, then for 9 min in a separate part of the extraction line with SAES AP-10 getters. They were analyzed with a MAP 215-50 mass spectrometer installed at the Geological Survey of Norway. The peaks were determined during peak hopping for 10 cycles (15 integrations pr. cycle) on the different masses ($^{41-35}$Ar) on a Balzers electron multiplier and regressed back to zero inlet time. Blanks were analyzed every 3rd measurement. After blank correction, a correction for mass fractionation, ^{37}Ar and ^{39}Ar decay and neutron-induced interference reactions produced in the reactor was undertaken using in-house software (AgeMonster written by Ganerød, 2010). It implements the equations of McDougall and Harrison (1999) and the newly proposed decay constant for ^{40}K after Renne et al. (2010). A ^{40}Ar/^{36}Ar ratio of 298.56 ± 0.31 from Lee et al. (2006) was used for the atmospheric argon correction and mass fractionation calculation (power law). We calculated *J*-values relative to an age of 28.619 ± 0.034 Ma for the TCR sanidine flux monitor (Renne et al., 2010). Weighted mean ages are calculated by weighting on the inverse of the variance (analytical uncertainties).

3.4. AF demagnetization

Previous studies on the Deccan Traps (Vandamme et al., 1991; Vandamme and Courtillot, 1992) have shown that alternating field (AF) demagnetization is more efficient as compared to thermal demagnetization. The AF-demagnetization experiments were done in the magnetic-laboratory Grubenhagen at the Leibniz Institute for Applied Geophysics (LIAG) Hannover. For demagnetization a Magnon, MI AFD 300demagnetizer was used. The remanence measurements were done with a superconducting magnetometer (2G Enterprises, Model: SRM 760-5.0 LC; Rolf, 2000) and a spinner magnetometer (Magnon, MI ***SM 1). After some test runs, the specimens were demagnetized with an average of 10 steps in fields up to 120 mT. In total 183 specimens were AF demagnetized. The analysis of the vector demagnetization diagrams (Zijderveld, 1967) was done by use of the principal component analysis after Kirschvink (1980). For analysis the Palmag (Version 3.2) package of Maier and Bachtadse (1994) and the Stereonet 8 (Version 8.8.9) (Allmendinger, 2013) software was used. In addition temperature dependent susceptibility measurements were done to identify the magnetic carriers.

4. ^{40}Ar–^{39}Ar age determination

As the age constraints of s41 and s46 are characterized by high uncertainties and are therefore not reliable, only the data for s32 are presented here. Two analyses for s32 were performed on plagioclase separates and whole rock. The results are displayed in Table 2 and illustrated in Fig. 7. The raw experimental data are available in Supplemental data file 1.

Both samples show concordant apparent ages over parts of the spectrum. The "big feldspar" sample (Fig. 7a) gives a weighted mean plateau age (WMPA) of 67.12 ± 0.44 Ma. The inverse isochron overlaps with the WMPA and the intercept is well constrained and overlaps with atmospheric value, hence, we interpret

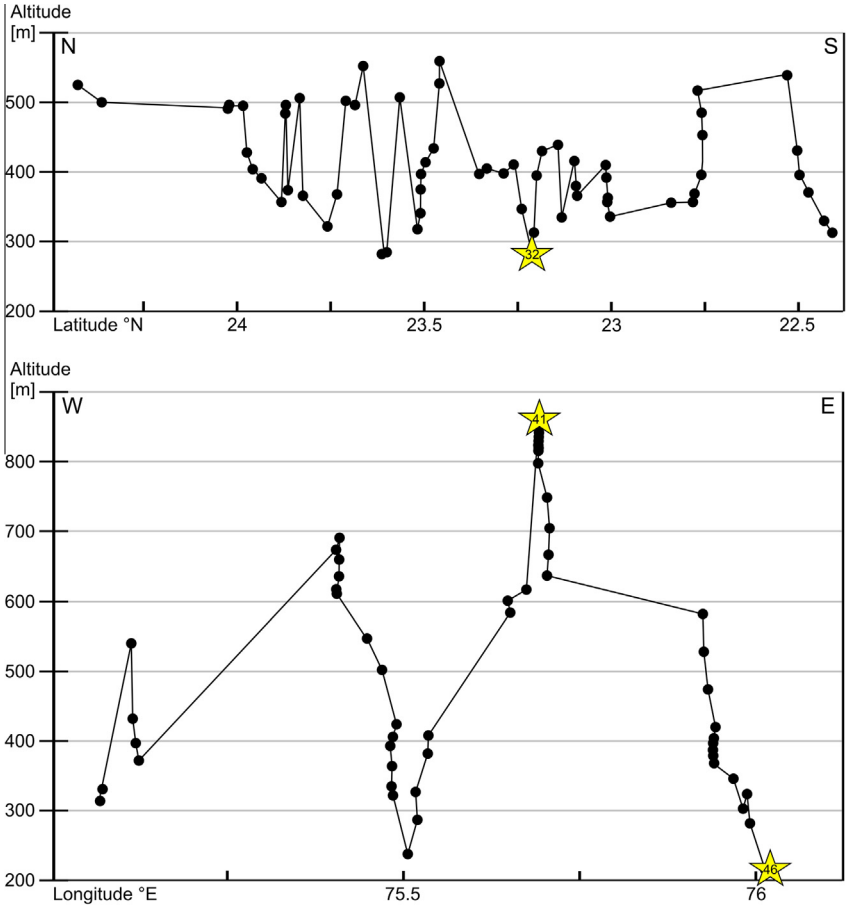


Fig. 6. Elevation vs. latitude/longitude for the sampled sites (filled circle) along the N–S and the W–E transects, respectively. The asterisks represent the sampling sites used for ^{40}Ar–^{39}Ar dating.

Table 2
$^{40}Ar/^{39}Ar$ results for CO_2 laser step heating experiments. Ages are in Ma. Uncertainties are reported as analytical errors.

Sample	Lat/long	Steps (n)	$\%^{39}Ar$	Spectrum analysis				Inverse isochron analysis			
				Age ± 1.96σ	MSWD (P)	TFA ± 1.96σ	K/Ca ± 1.96σ	Age ± 1.96σ	MSWD (P)	Intercept ± 1.96σ	Spread%
$s32^{P}$	23.2111/74.4559	3–12 (10)	96.89	67.12 ± 0.44	1.013 (0.426)	67.04 ± 0.6	0.092 ± 0.00054	67.3 ± 0.63	1.083 (0.371)	297.56 ± 2.62	68.7
$s32^{WR}$	23.2111/74.4559	4–10 (7)	64.9	67.73 ± 0.22	1.872 (0.081)	69.45 ± 0.3	0.35 ± 0.0019	67.33 ± 0.82	18.69 (0.0096)	312.71 ± 28.33	11.3

the WMPA to be the best age estimate for this sample. The whole rock sample (Fig. 7b) shows a saddle shaped spectrum, indicative of excess argon, however the uncertainty of the intercept of the inverse isochron analysis cannot resolve this fully. For this sample we calculate a WMPA of 67.73 ± 0.22, however, this age is not coherent with the "big feldspar" age, probably due to excess argon in the plateau age. Given the fact that the inverse isochron ages overlap, we conclude that the WMPA of 67.12 ± 0.44 Ma is the best estimate of the crystallization age of sample s32.

5. Magnetic studies

5.1. Demagnetization characteristics

The Curie/Néel (T_C/T_N) temperature is characteristic for every mineral and the contributing magnetic phases can be inferred from the drop in the bulk susceptibility at that temperature. Analysis of the temperature dependent susceptibility indicates that the main magnetic carrier is titanomagnetite with variable content of titanium. Some representative temperature dependent susceptibility curves are shown in Fig. 8.

NRM intensities vary between 0.18 A/m and 33.2 A/m and have an overall median of 5.2 A/m. The samples generally bear a stable remanent magnetization, which could be isolated after the AF-demagnetization in peak fields of $\geqslant$10 mT. The residual magnetization of the initial NRM at peak fields of 100 mT varies between 0.1% and 36.4%. Specimens showing a very high residual magnetization are either dominated by single domain (SD)-grains or/and contain high coercive minerals, like hematite or goethite. However, the average of 7% residual magnetization shows that the specimens are mostly demagnetized at 100 mT. Fig. 9 summarizes the demagnetization curves. The diagrams also include ideal AF-curves of SD-, PSD- and MD-magnetite grains as suggested by Palmer et al. (1996). MD-grain controlled curves show a steep decline in intensity even if a low AF-field is applied. The curves for PSD-particle dominated specimens are similar, but the decay of the magnetization is decelerated. The shape of the curves changes from concave to convex when SD-grains embody the prevailing magnetic domain size. Specimens with high SD-content usually have a hard magnetization and require a high coercive force for demagnetization. A

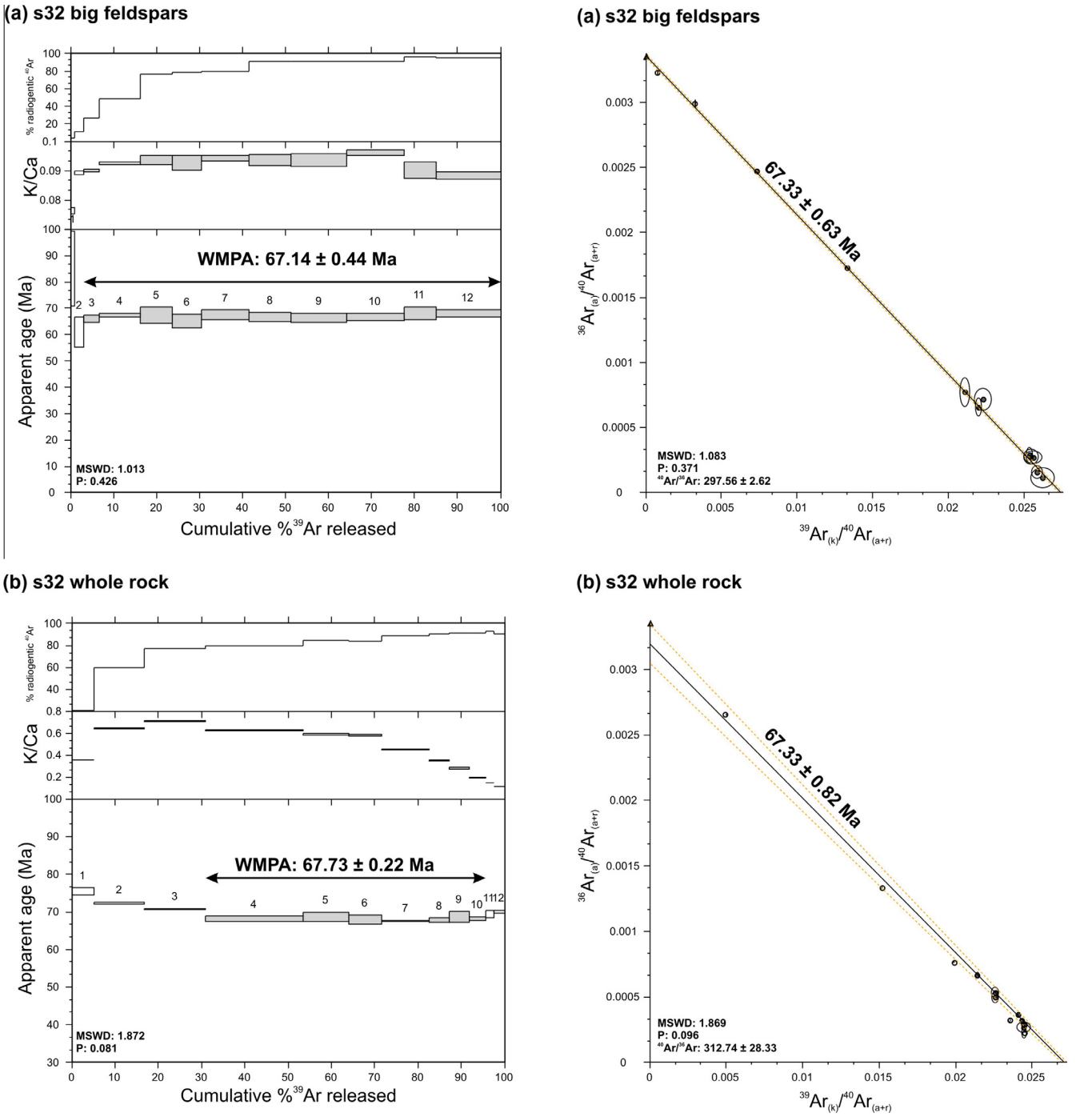


Fig. 7. ^{40}Ar–^{39}Ar age spectra and K/Ca ratios (above) with inverse isochron results shown on the right side, for the two CO_2 laser step heating experiments from s32. The error bars on each step, and for the calculated dates are shown at the 1.96σ level. The numbers above each bar resembles the experiment number (step) in A.1.

vast majority of AF-curves exhibit the characteristics for MD- and PSD-grains, while a few curves are whaleback-shaped and indicate a high fraction of SD-grains. The mean destructive fields (MDF, a proxy for the coercivity of the magnetic population) reveal a broad scatter (MDF: 3–73 mT). The scatter can be explained by variation in the domain size of the magnetic grains or it may be due to the presence of high coercive minerals, like (titano-) hematite or goethite. The calculated Koenigsberger Q-ratio (Koenigsberger, 1936) in majority cases ranges between 1 and 10 (median 4.99), a common array for basalt. Only few specimens show a Q-ratio higher than 15 including three specimens with more than 40 and one more than 50. Such high Q ratios need careful attention as their magnetization may be have been affected by lightning. These suspected lightning struck specimens were discarded from further evaluation.

5.2. Zijderveld diagrams

Typical demagnetization behavior for characteristic specimens is shown in vector-diagrams (Fig. 10), the Zijderveld-plots (Zijderveld, 1967). A secondary low coercive direction isolated between 5 and 10 mT in the direction of the present day Earth's

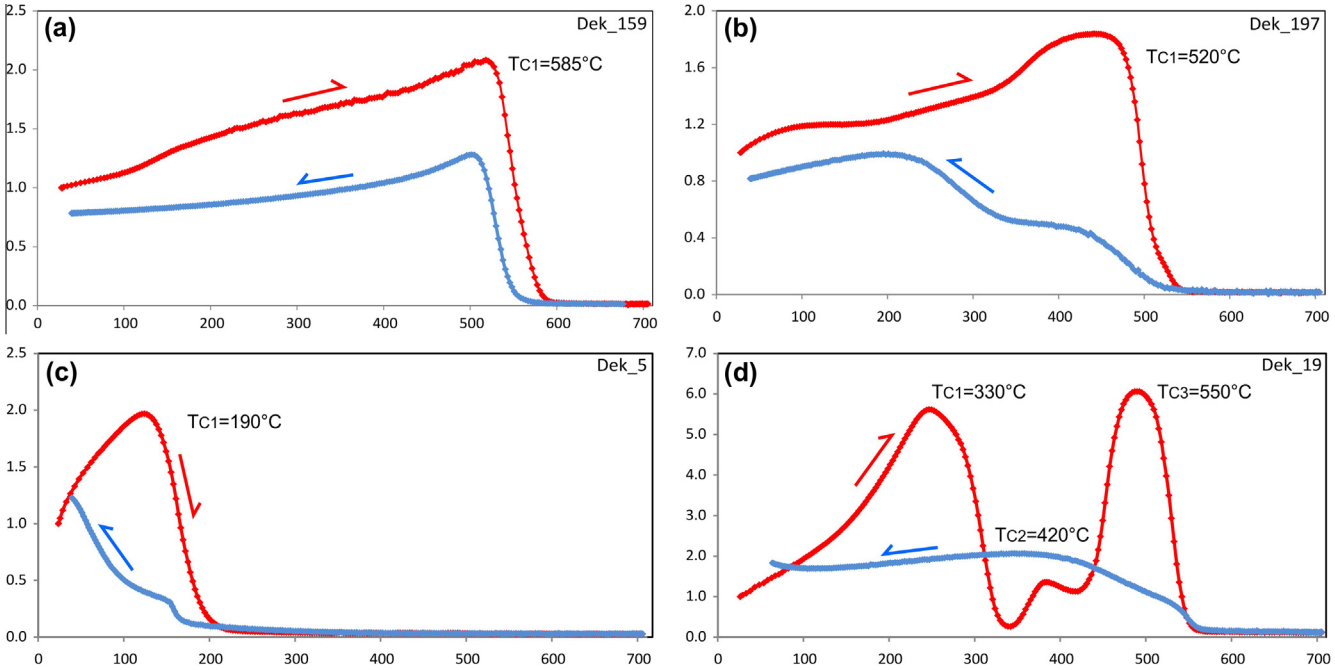


Fig. 8. Temperature dependent susceptibility curves for some representative tholeiitic basalt samples from the Malwa Plateau. The magnetic behavior of the samples is dominated by (a) magnetite, (b) titanomagnetite with low titanium, (c) titanomagnetite with high titanium and (d) multiple phases of titanomagnetite.

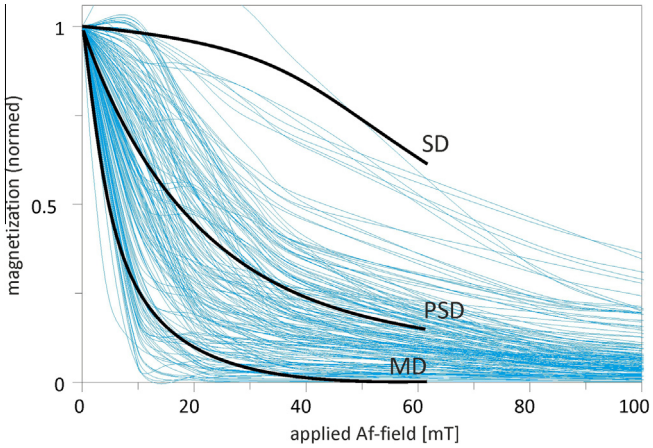


Fig. 9. AF-demagnetization curves for all measured specimen. The SD/PSD/MD are according to Palmer et al. (1996).

magnetic field is visible in most of the specimens. After the removal of this component the original magnetization (characteristic component) and sometimes other secondary components are still present. The cylinders 24-2 and 21-1 are examples of specimens showing a single characteristic component with normal polarity. The characteristic direction of magnetic component heads to the diagram center and reaches the origin after an AF of 80 mT. This is a typical behavior of (titano-)magnetite-bearing samples. Sample 174-1-1 shows a similar direction of the characteristic component but also an influence of a second component. However, after the demagnetization step of 40 mT the influence of this second component vanishes. The residual magnetization is hard and a field of 100 mT is insufficient to remove the magnetization. This hard remnant magnetization indicates a small fraction of high coercive (titano-)hematite. The specimens 35-2 and 47-1 possess a reversed polarity while 35-2 shows only a slight curvature after the removal of a viscous magnetization. Specimen 47-1 shows a strong second component, removed at a field of 40 mT. Some specimens are not suitable to record information of the ancient magnetic field as there is no confident result obtained after the removal of the secondary components. Specimen 44-2 is shown as an example for these poor quality samples.

5.3. Site groups

As most of the analyzed cylinders show a stable and well defined main magnetic component, the characteristic direction shows a low median absolute deviation (MA; Kirschvink, 1980) with a median of 1.9 for all specimens. For most specimens an AF of 10 mT removed the viscous magnetization and the 10–100 mT segment (M010–M100) was used for the calculation of the characteristic magnetization. In cases with a strong second component, the M040–M100 segment could be used to obtain the reliable direction of the main magnetic component.

The paleomagnetic data set is compiled in Table 3 and shown in stereographic projections in Fig. 13. For some sites, only two cylinders were analyzed for remanence characteristics. This low quantity of specimens is acceptable, as most specimens show a well-defined and reliable remanence vector. Sampling sites were grouped as "site groups" (sg) based on their geographic location, altitude and magnetic polarity. The location of the site groups can be seen in Fig. 11. For all sites (and site groups) the mean directions were calculated using Fisher (1953) statistics. Site groups 1, 2, 8 are well defined and show a distinct cluster of the site directions ($\alpha 95_{sg1}$: 7.8; $\alpha 95_{sg2}$: 15.6; $\alpha 95_{sg8}$: 14.3). Sg 6 and sg 7 do not show a precise grouping ($\alpha 95_{sg6}$: 100; $\alpha 95_{sg7}$: 54.7) which is also due to the low quantity of data. However, the magnetic polarity in all sites and site groups can be clearly identified as normal and reverse, respectively. Site group 9 ($\alpha 95_{sg9}$: 22) shows a steep relatively weak remanent magnetization. As this site is elevated from the surrounding area, the magnetic remanence is suspicious to be induced by lightning strikes, even though the Koenigerbergers factors are quite moderate for most of the samples.

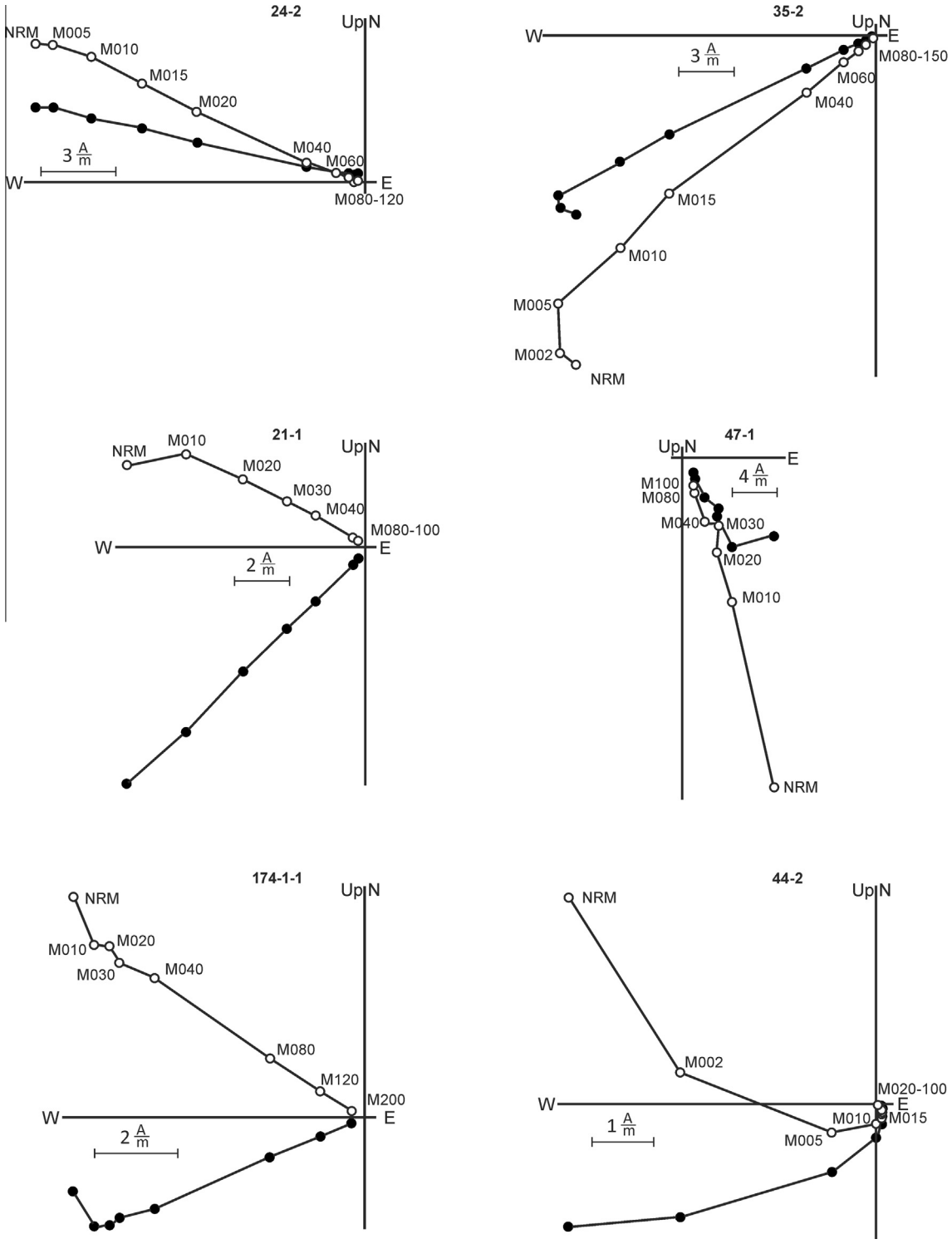


Fig. 10. Six examples of orthogonal projection of alternating field demagnetizations of representative samples. Filled circles in orthogonal projections correspond to the N–S/E–W planes, open circles to the N–S vertical planes.

5.4. Magnetostratigraphy

The magnetostratigraphic polarity, the position and altitude of samples were used for construction of three profiles across the western and southern rims of the Malwa Plateau (Fig. 12).

In the northern part of the plateau, the profile A–A′ indicates a relatively constant thickness of about 110 m for the lava flow series which poured out during the magnetochron C30n. The trace of C30n/C29r isochron in this section indicates a slight tilt in northern and southern directions respectively, which is less than 1°. At Pratapgarh the contact between the Deccan flows and the underlying Vindhyan sandstones is at an altitude of 300 m a.m.s.l., the C30n/C29r isochron at an altitude of 400 m and the main plateau surface at an average altitude of 500 m. In the southern sector of this profile (A′), a displacement along a normal fault in the west drops the sequence by 60 m.

Table 3
Paleomagnetic results for the Malwa lava flows. sg: Grouping based on their geographic location, altitude and magnetic polarity (see text); *n*/*N*: number of specimens (samples) used in the statistics/number of specimen (samples) measured; NRM: intensity of natural remanent magnetization (mA/m); D° (PCA)/I° (PCA): declination/inclination of principal component analysis in geographic coordinates; *k* and $\alpha 95$: parameters of Fisher statistics.

Site	sg	*n*/*N*	Altitude (m)	NRM (mA/m)	mdf (mT)	Q	D° (PCA)	I° (PCA)	κ	$\alpha 95/\sigma$	Site lat.	Site long.
s7	1	2/2	402	6.02E+03	8	4.1	356	−30	672.9	6.8	23.9623	74.7242
s17	1	2/2	320	1.13E+04	24	14.5	333	−28	63.1	22.4	23.7613	74.6352
s19	1	5/8	283	9.39E+03	21	8.02	348	−28	73.6	8	23.6019	74.4884
s21	1	2/2	281	5.02E+03	14	6.82	346	−21	178	13.3	23.6085	74.4945
s24	1	3/4	364	4.99E+03	23	16.6	347	−23.4	113.8	9.5	23.8272	74.6357
s25	1	0/2	372	1.33E+03	7	1.52	355	−29	8.6	64.7	23.8676	74.6752
s107	1	0/2	316	1.12E+04	14	11.4	323	−18.3	3.7	111.2	23.5192	74.5978
SG1		5/7					346	−26	77.7	7.8		
s5	2	2/2	493	1.33E+04	14	12.7	147	50	246.1	11.3	24.0256	74.783
s6	2	4/4	426	1.10E+04	9	9.52	164	16	103.1	7.9	23.978	74.7414
s10	2	2/2	493	5.66E+03	7	4.6	126	43	371.2	9.2	23.9883	74.7491
s12	2	2/2	482	8.33E+03	17	8.35	134	51	294.2	10.3	23.8747	74.7876
s14	2	2/3	550	1.43E+04	15	18	164	19	91.9	18.5	23.6651	74.8244
s15	2	3/3	500	6.95E+03	7	14	162	58	2717	1.9	23.7119	74.7158
s105	2	2/2	373	7.39E+03	21	9.53	151	28	76.6	20.3	23.5105	74.6807
s106	2	0/2	339	2.23E+04	22	24.7	153	1	13.7	49.7	23.5109	74.6718
SG2		7/8					151	39	13.7	15.6		
s32	3	3/3	281	7.42E+03	13	4.09	332	−8.3	147.4	8.3	23.2111	74.4559
s33	3	3/3	345	6.89E+03	27	11.5	300	−58.4	378.1	5.2	23.2387	74.4775
s99	3	2/2	333	7.15E+03	62	9.65	327	−23.5	729.6	6.5	23.131	74.5019
s111	3	2/2	361	7.99E+02	12	0.67	7	−45.5	2215	3.8	23.007	74.6681
s113	3	2/2	364	7.53E+03	33	13.7	316	−6.5	507.1	7.9	23.0896	74.5363
SG3		5/5					329	−30	6.1	29.7		
s28	4	2/2	437	4.25E+03	11	8.58	127	66	1514	2.6	23.1411	74.5617
s29	4	2/2	428	2.72E+04	14	36.4	183	9	42	27.6	23.184	74.5519
s30	4	2/2	393	1.67E+04	20	16.3	157	30	234.6	11.6	23.1985	74.5256
s35	4	2/2	396	3.42E+03	5	1.82	158.4	31.1	87.1	19	23.2873	74.4999
s36	4	2/2	403	7.51E+03	25	4.07	154	16	104.1	17.4	23.3326	74.4902
s37	4	2/2	394	9.39E+03	23	4.59	151	55	709.2	6.6	23.3531	74.4559
s109	4	0/2	390	1.30E+04	7	11.9	139	17	3.2	127.8	23.0109	74.6855
s114	4	2/2	415	6.19E+03	26	5.91	143	8	3.2	–	23.097	74.5363
SG4		7/8					156	31	8.4	20.3		
s83-5	5	3/3	328	1.62E+03	15	3.12	325	−21	24.3	20.7	22.4031	75.0689
s93-5	5	3/3	367	5.63E+03	12	7.1	315	−14	78.2	11.4	22.7554	74.7472
SG5		2/2					320	−17.6	47	26		
s87	6	2/2	430	5.88E+03	8	4.61	114	75	126.8	15.7	22.4976	75.1165
s88_91	6	2/2	495	6.91E+03	11	6.47	155	17	23.2	37.5	22.5246	75.1145
SG6		2/2					147	47	1.7	100		
s55	7	2/2	402	7.11E+03	67	7.1	3	−63	1122	5.3	22.4636	75.9341
s64	7	2/2	380	1.20E+03	25	1.24	334	−59	1900	4.1	22.3769	75.5316
s65	7	2/2	325	1.28E+04	5	13.9	257.5	−1.5	221	38.5	22.3586	75.5145
s70	7	2/2	362	1.23E+03	8	2.33	327	−48	818	6.2	22.343	75.4813
SG7		4/4					309	−52	2.8	54.7		
s43	8	2/2	615	1.23E+04	21	7.58	151	40	28.3	33.8	22.4665	75.6703
s44	8	0/2	599	7.00E+03	12	7.23	155	32	7.4	70.3	22.455	75.6441
s45	8	0/2	582	8.44E+02	32	0.77	179	40	139.6	15	22.4608	75.6476
s56	8	2/2	418	2.85E+03	6	1.42	130	56	402.6	8.8	22.465	75.9364
s63	8	2/2	406	4.80E+03	14	1.48	119	57	236.8	11.5	22.3818	75.5325
s72	8	2/2	404	4.79E+03	6	4	145	75	70.5	21	22.3829	75.4825
s73–75	8	2/2	484	4.67E+03	7	2.87	155	60	146.2	14.7	22.4147	75.4876
SG8		5/7					140	58	23.8	14.3		
s41a	9	11/11	858	2.34E+03	27	2.92	198	45	103.5	4.3	22.4608	75.688
s41b	9	2/2	849	1.47E+03	5	1.24	253	69	17.9	43	22.4614	75.6881
s41c	9	0/3	845	2.23E+04	6	30.4	44	−7	10.7	31.8	22.4628	75.6878
s41d	9	5/6	837	2.04E+03	6	3.36	228	56	8.7	24.2	22.4634	75.6875
s41e	9	3/3	828	1.43E+03	11	0.52	280	89	436.7	4.8	22.4643	75.6873
s41f	9	9/9	820	2.59E+03	17	1.78	168	54	91.5	5.1	22.4649	75.6873
s41g	9	0/3	814	1.99E+03	7	1.63	175	−16.7	188.4	7.3	22.4651	75.6869
SG9		5/7					207	66	10.5	22		

In the very west of the plateau (profile B–B′) the lava series runs out. The contact of the basement is at an altitude of 300 m and the reduced thickness of the C30n sequence (~60 m) indicates that initially fewer lava flows reached this part of the Malwa. Towards the eastern part of the plateau the rise of the C30n/C29r isochron suggests a normal fault, which may be related to the fault system close to the A–A′ profile.

The southern profile (C–C′) covers most of the western part of the southern Malwa Plateau. The C30n/C29r isochron can be traced at a constant altitude of 400 m, indicating that there

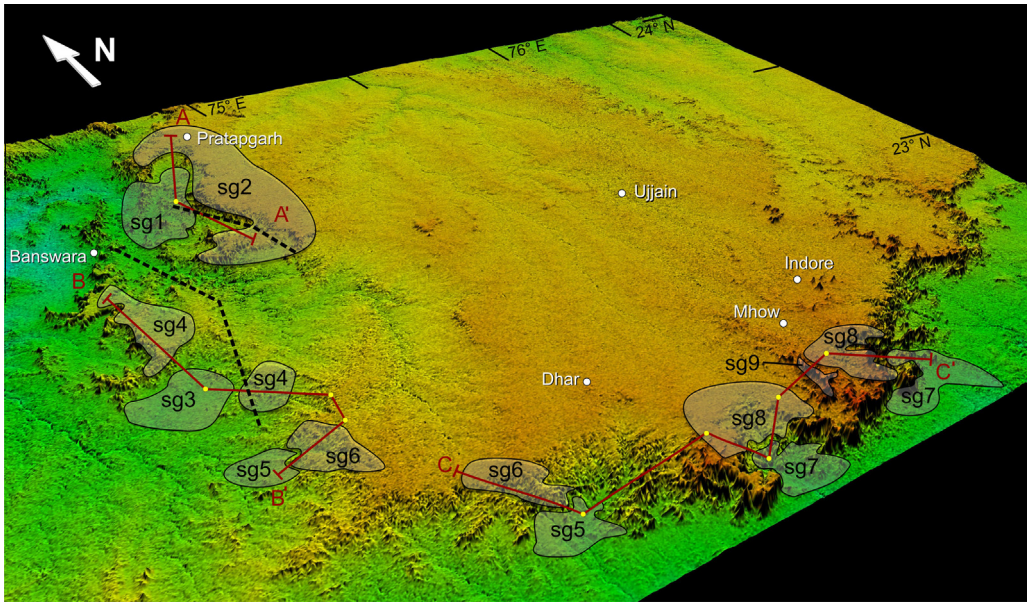


Fig. 11. 3-Dimensional digital elevation model (DEM) of the western Malwa Plateau showing the course of the profiles and the location of the site groups (sg). Data for DEM was provided by the ASTER G-DEM project (METI and NASA).

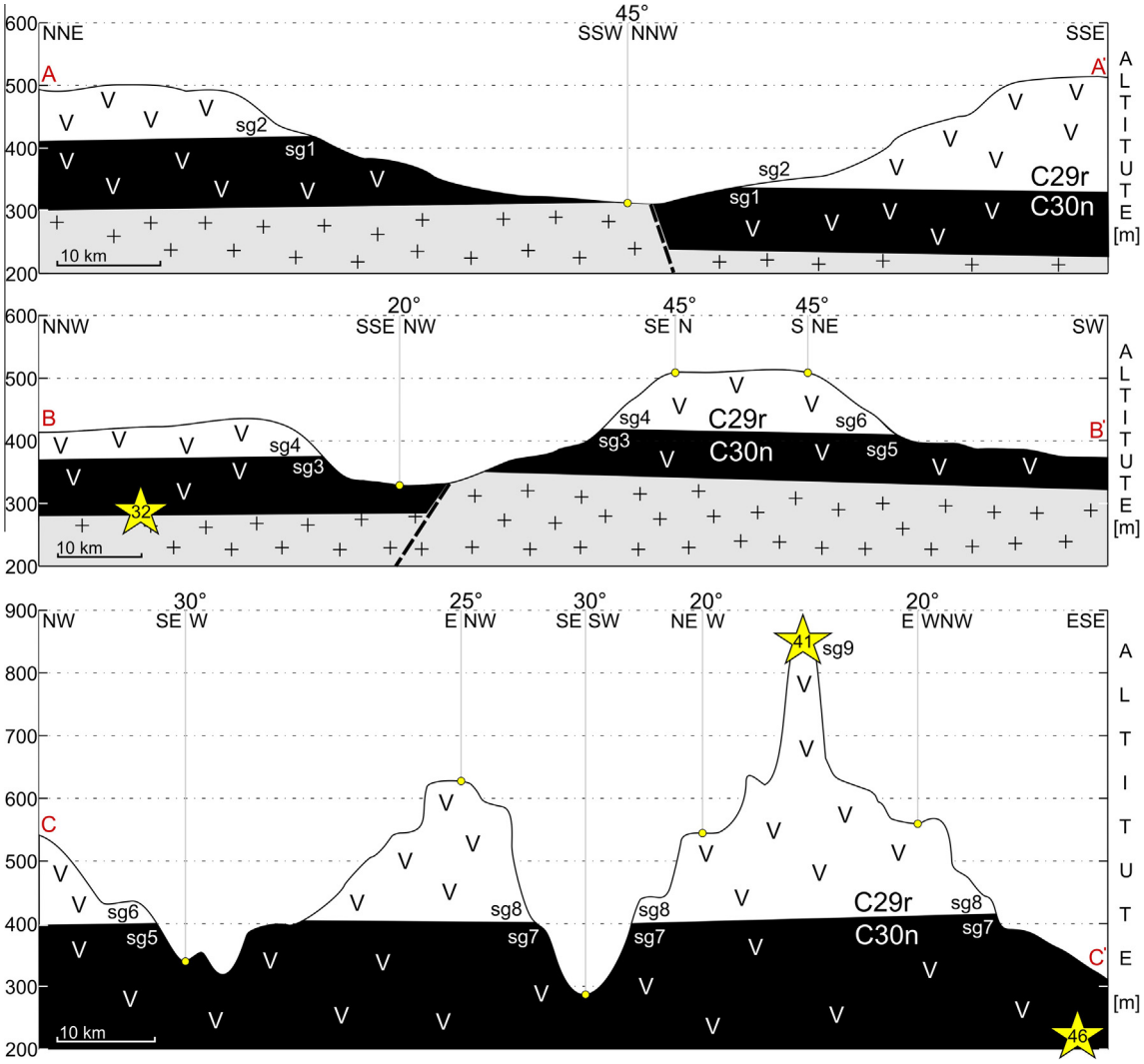


Fig. 12. Magnetostratigraphic profiles along the western/southern rims of the Malwa Plateau. For course of the profiles see Fig. 11. Asterisks mark the sites used for radiometric ^{40}Ar–^{39}Ar dating.

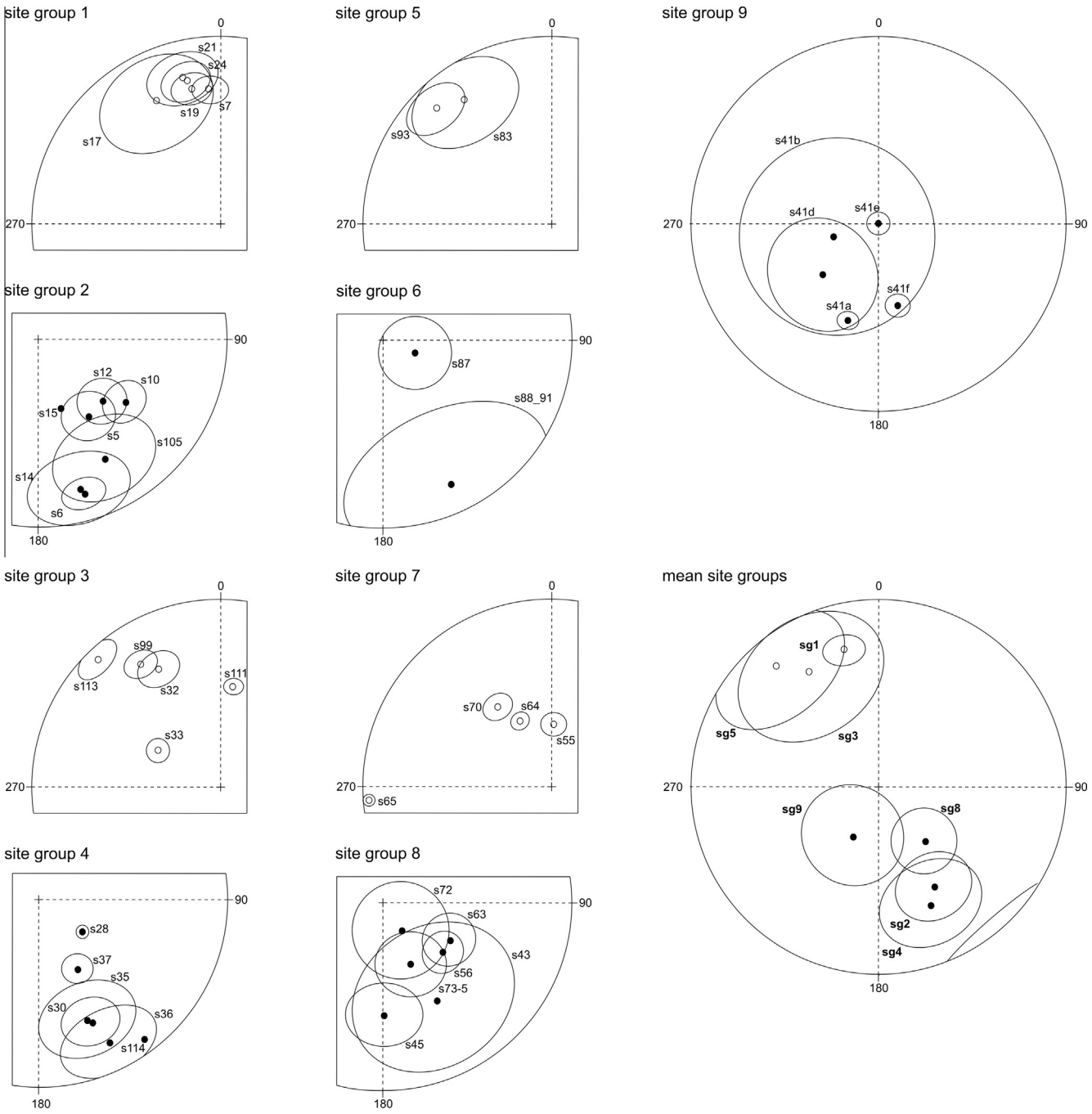


Fig. 13. Equal-area (stereographic) projections of site mean directions and $\alpha 95$ of the respective site groups. The lower right projection shows the mean directions and $\alpha 95$ for the site groups. Open circles in upper hemisphere (normal magnetized) and filled circles in lower hemisphere (reverse magnetized).

are no major normal or reversed faults in this area. As the contact with the underlying basement is covered by alluvium from the Narmada River, the thickness of the lava pile can only be indirectly estimated. However, Khadri (2003) recorded normal polarity flows close to the Narmada River at an altitude of 160 m. This leads to a minimum thickness of 240 m for lava flows extruded during the magnetochron C30n. The average altitude of the plateau in this southern region is 550 m which is slightly higher than in the northern area, where the plateau has an average elevation of 500 m above a.m.s.l. Southwest of Mhow the Singachori hill with 860 m a.m.s.l. represents the highest peak of the Malwa region. The C30n/C29r in this section is recorded at 400 m. The lava sequence erupted during the magnetochron C29r has a minimum thickness of 460 m. These sums up to a cumulative thickness of at least 710 m for the lava pile. On the very top of Singachori hill, Khadri (2003) reported a polarity transition from normal to reverse (N–R transition). Our samples from this site (sg 9) do not show a clear N–R transition. More likely, the highest flows of this site resemble a geomagnetic excursion or the beginning of an Earth's magnetic field reversal. However, the magnetic remanence can also be characterized by lightning strike magnetization as it is indicated by the steep direction of the remanence vector and a regional increased *Q*-factor (up to 30) at some sites.

6. Interpretations and discussion

6.1. Extrusion history of the Malwa flows

In the lava pile of the Malwa Plateau a transition from normal to reversed magnetization has been determined and could be mapped in NS and E–W sections. The ^{40}Ar–^{39}Ar ages allow a correlation to the C30n/C29r transition of the geomagnetic polarity time scale (Gradstein et al., 2012). The age for site 32 with 67.12 ± 0.44 Ma can be clearly correlated with the magnetochron C30n. The overlying transition can therefore be linked to the C30n/C29r (66.4 Ma: Gradstein et al., 2012) transition. This finding differs from the earlier correlations by Khadri (2003) who relates the N–R transition of the Malwa to the C34n/C33r transition, which may be due to lack of reliable age data.

The structure of the plateau suggests that the eruption center is located in the south of the plateau. The thickness of the lava flow sequence thins out in the western and northern districts of the Malwa. Not only the total thickness but also the amount of lava which poured out during the magnetochron C30n is highest in the southern part of the province. It is likely, that the main pulse of the eruption extruded close to Singachori Hill, where the lava sequence reaches its maximum thickness. This is also supported by the data of Schöbel and de Wall (2014), who determined bipolar directions of the lava flow axes using the magnetic fabric (anisotropy of magnetic susceptibility) of the tholeiitic flows. Fig. 14 shows interpreted flow directions for the same study area. The results from the magnetic fabric study fit well with the magnetostratigraphic structure derived from this study. All observations favor the Narmada-Tapi rift zone as a possible source for the Malwa basalts.

Narmada- and Tapi-Lineaments form a graben system with the central Satpura mountains as the central horst. The Narmada Tapi rift zone is one of the most prominent rift zones of India and has been targeted for various geophysical surveys (e.g. Kaila, 1986; Nayak, 1990; Ravi Shankar, 1991; Verma and Banerjee, 1992; Sridhar and Tewari, 2001). The Narmada Tapi zone is characterized by a discontinuous median positive gravity anomaly (up to 40 mgal), superimposed by isolated gravity highs/lows, aeromagnetic, seismic and heat flow anomalies (Bhattacharji et al., 1996). This rift system has a long-term geological history with several events of reactivation (for a review see Chamyal et al., 2002). The graben structure was reactivated during the Late Cretaceous (Agarwal et al., 1994) and is therefore covered by the Cretaceous Lameta beds and marine sediments (Bagh beds) in the western part of the Narmada valley. While the rifting continued the Deccan flows erupted and covered the graben and the surrounding area. Also Intertrappean beds can be found in the sediments of the Narmada valley (e.g. Blanford, 1869; Pascoe, 1964; Tandon et al., 1995).

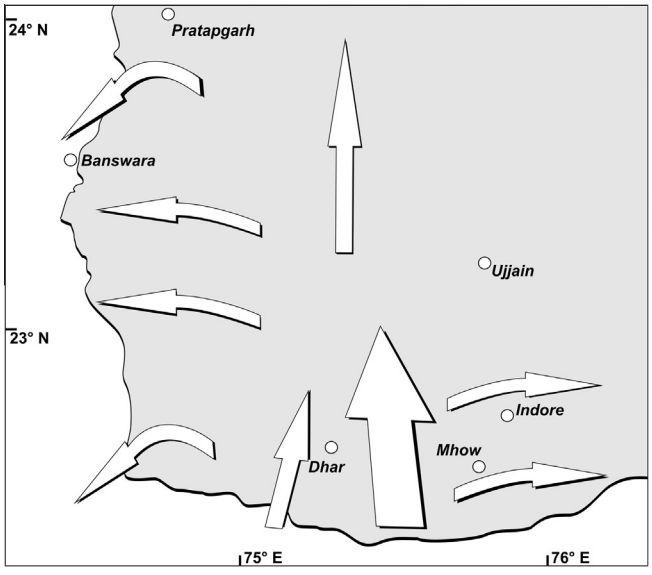


Fig. 14. Flow pattern of the Malwa. Derived from magnetic fabric analysis (modified after Schöbel and de Wall, submitted for publication).

6.2. Comparison with the other Northern Deccan sequences

A correlation of Malwa, Mandla, Toranmal lavasw (Northern Deccan Province) is not trivial. However, it can be assumed that the Mandla lava flows fall in the same age bracket as the Malwa flows, based on close geochemical similarity (Peng et al., 1998) and the presence of the C30n/C29r transition (e.g. Poornachandra Rao and Bhalla, 1981). Allègre et al. (1999) estimated an average age of 67.2 ± 1.0 Ma for the Mandla area using the ^{187}Re–^{187}Os systematics, which is in good agreement with the age data for the Malwa ages obtained in this study. As the Mandla lobe is also situated along the Narmada-Tapi rift zone, it seems reasonable to assume that the Mandla flows also erupted from this rift zone. In the east of the Mandla area, at the Amarkantak–Dindori tract, an R–N transition in a 300 m thick section is reported by Athavale (1970). As this transition is in the top of the sequence it is likely that it represents the (C29r/C29n) transition. The (C29r/C29n) transition often occurs in the southwestern series and is also usually on the top of the sequences. This would indicate that the Mandla flows erupted mainly synchronous with the southwestern lava flows. However, the volume and eruption rates were much lower in the Mandla area.

The Toranmal section, situated within the Narmada-Tapi rift zone, is presumably younger as there are mainly reversed polarized lava flows and a clear N-R transition is not recognizable. Geochemically the Toranmal flows are very similar to the upper Malwa flows, as both show Poladpur-like compositions (Peng et al., 1998; Mahoney et al., 2000).

6.3. Comparison with the Western Ghats

Linkage between Malwa flows and the main profiles of the Western Ghats is possible on account of precise ^{40}Ar–^{39}Ar ages. As the C30n/C29r transition usually does not occur in the lava flows of the Western Ghats, the Malwa flows erupted prior to the Western Ghats series. However, the C30n/C29r transition falls within the Jawhar Fm of the Kalsubai Supergroup, the basal formation of the Western Ghats. Therefore, the gap between first lavas of the Malwa and the onset of the Western Ghats series cannot be significantly longer than 1 Ma or is even shorter. As the Malwa eruptions go far into the magnetochron C29r, it is likely that the Western Ghats and the Northern Deccan Province erupted synchronously. A time gap of 2.5 Ma from the end of the Northern Deccan Volcanism to the start of Southwestern Flows, as is has been proposed before (e.g. Chenet et al., 2007; Keller et al., 2009a and references therein) seems therefore too long. Their calculated quiescence of 2.5 Ma results from the assumption that the start of the northern Deccan activity can be linked to the C30r/C30n transition and the termination in the beginning of the C30n. At least in the Malwa Plateau these assumptions contradict with the actual setting. Fig. 15 shows a magneto-stratigraphic correlation of the Northern Deccan Province segments to some main profiles of the Western Ghats. The main eruption activity during the C29r was not restricted to the Western Ghats and can also be assumed for the Northern Deccan Province. This narrows the time span for eruption of the LIP, as the main activity in the northern and southern parts of the Deccan

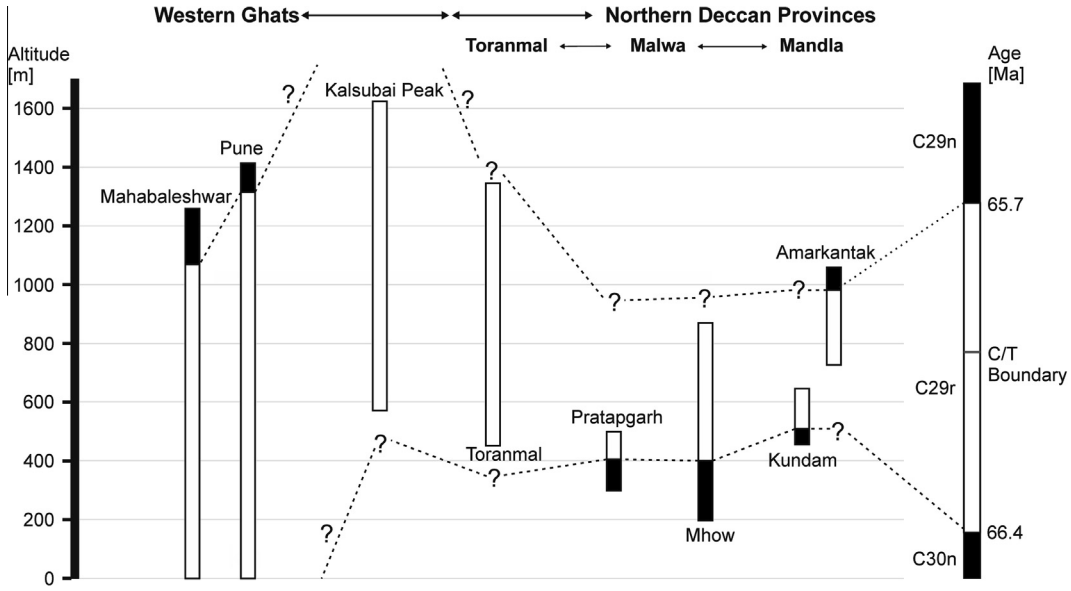


Fig. 15. Magneto-stratigraphic correlation of the Northern Deccan Province segments to some southern profiles. Data from Courtillot et al. (1986), Poornachandra Rao and Bhalla (1981) and Sreenivasa Rao et al. (1985).

LIP was during the 0.7 Ma duration magnetochron C29r. As the duration of the volcanic event is crucial for the impact on the atmosphere, this short but voluminous eruption phase favors a strong influence of the Deccan flows on the prevalent global climate.

6.4. Proposed geochemical stratigraphy for the Northern Deccan Province

The Malwa lavas started erupting during the magnetochron C30n and are partly synchronous to the Kalsubai Supergroup. However, the source areas for the Malwa and the Western Ghats series are several hundred km apart and each site would show its own geochemical evolution. Therefore the Malwa flows and also the other segments of Northern Deccan Province cannot be directly correlated with the geochemical stratigraphy of the Western Ghats. The Toranmal section shows the frontiers of geochemical correlation. Mahoney et al. (2000) identified flows in the Toranmal section which show similar geochemical fingerprints to the formations of the Western Ghats (Poladpur, Bushe and Mahabaleshwar). Therefore they tried to correlate these flows to the southwestern geochemical stratigraphy but failed to link the sections, as the flows of the Toranmal section show a stratigraphic order clearly distinct from the flows of Western Ghats series. This demonstrates that the lava flows of the Northern Province have no geochemical stratigraphic relationship with the southwestern sequence despite similarities in their geochemical composition. A more confident correlation can be done using the geomagnetic field transitions, which have to be dated by precise radiometric methods. However, Peng et al. (1998) and Mahoney et al. (2000) show that the Northern Deccan Sequences (mainly Malwa, Mandla, Toranmal but also Chikhaldara) show similar geochemical characteristics. Also the sites display a significant geochemical evolution in their stratigraphy. It is evident that there is a need for a new geochemical stratigraphy, incorporating the geochemical characteristics of the Northern Deccan Province. Sreenivasa Rao et al. (1985) introduced the term "Narmada Fm" to describe the northern flows of the Deccan LIP. We suggest to extend the "Narmada Fm" to the "Narmada Subgroup". A geochemical characterization of the respective formations of the "Narmada Subgroup" is under progress.

6.5. Estimation of volume and eruption rate of lava

Not many geophysical studies have been done in the Malwa area up to now. Kaila (1986) derived a shallow depth section from seismic reflections in the Malwa area. The profile starts at Ujjain in the north, continues to the south through Indore, crosses the Narmada and Tapi rivers and ends further in the south. According to the profile the Malwa lava flows have a constant thickness of 100–200 m in the Ujjain-Indore transect, which increases significantly up to 400–500 m, toward the south of Indore. The results of the reflection seismic studies support our stratigraphic findings. For the calculation of the present dimension of the Malwa traps we assume an average thickness of 300 m for the central parts and a mean thickness of 150 m for the lateral areas of the plateau. For a 250 km long and 20 km wide W–E strip from Dhar to Bhopal we estimate a lava pile thickness of 500 m. According to these assumptions the Malwa flood basalts presently cover an area of 80,170 km^2 with an estimated volume of 21,000 km^3. The initial volume was undoubtedly far greater. The mounts in the south of the plateau, for example indicate that these hills are only relicts of an extended lava sequence.

For an estimation of the eruption rate of the Malwa region, the lava volume and timing of activity during the C30n can be used. The precise age at the bottom of the sequence at site 32 (67.12 ± 0.44) constrains the lower age limit and the C30n/C29r transition can be used as upper age limit. We use the recently revised age of 66.4 Ma for the C30n/C29r transition (Gradstein et al., 2012). This is slightly older than the previous age of 65.6 Ma by Cande and Kent (1995) used for interpretations of the Western Ghats sections by Jay and Widdowson (2008). The time interval between eruption of the lowermost Malwa flows to the C30n/C29r transition lasted at least 0.3 to a maximum of 1.2 Ma. We can estimate the volume according to the presently existing C30n flows. As they are underlying the C29r flows, they may not have been significantly weathered. We therefore estimated an initial volume of 15,000 km^3 for the lava flows erupted during the C30n in the Malwa area. This leads to possible extrusion rates of 0.01–0.05 km^3/a for the Malwa area. Compared to the estimated 2.1–8.2 km^3/a eruption rates for the whole Deccan LIP (Coffin and Eldholm, 1993) the rate for the Malwa is low but seems

realistic being within the range of eruption rates assessed for other LIPs (Siberian Traps: >1.3 km^3/a, Renne and Basu, 1991; Columbia River Basalts 0.1–0.9 km^3/a, Coffin and Eldholm, 1993). However the ^{40}Ar–^{39}Ar site 32 lies in the west of the plateau, where the thickness of the plateau is significantly reduced. Therefore uncertainties remain whether these flows really resemble the first flows of the Malwa.

As most of the Malwa flows erupted during the magnetochron C29r it is likely that the eruptions lasted at least 0.4 Ma during C29r. It is assumable, that the Narmada-Tapi rift zone was still active and outpouring the stratigraphically higher lava flows during the K/T boundary (65.95 Ma: Kuiper et al., 2008). The duration for the Malwa flows can be estimated as a minimum of 0.7 Ma (C30n: 0.3 Ma; C29r: 0.4 Ma) and a maximum of 2.3 Ma (C30n: 1.6 Ma; C29r: 0.7 Ma). The initial volume of the Malwa flood basalt can be calculated from the duration of eruption and the eruption rate, which results in a minimum volume of 7000 km^3 and a maximum volume of 115,000 km^3. By using a moderate eruption rate of 0.03 km^3/a and a likely total eruption duration of 1.4 Ma (C30n: 0.9 Ma; C29r: 0.5 Ma) the initial volume of the Malwa flows multiples to 42,000 km^3, which seems to be realistic. However, compared to the voluminous provinces of the Western Ghats (e.g. Self et al., 2006), the Malwa flows play a volumetric minor but stratigraphic important role in the history of the Deccan LIP.

7. Conclusions

- The Malwa flows presently cover an area of about 80,000 km^2 and are therefore the largest segment district of the Northern Deccan Province. With a present volume of about 14,700 km^3 and an estimated initial volume of 28,000 km^3 the Malwa flows form an extensive but relatively thin sequence. The duration of eruption can be estimated between 0.6 Ma and 1.9 Ma.
- Magnetic and radiometric results suggest that the Malwa flows erupted during the magnetochrons C30n and C29r. Further rock magnetic data (Schöbel and de Wall, 2014) and the internal structure of the plateau suggest that the Malwa flows most likely erupted from the Narmada-Tapi rift zone.
- A magnetostratigraphic, radiometric and geochemical correlation of the Malwa, Mandla and Toranmal section seems possible. The basal flows of the Northern Deccan Province started to erupt earlier than the lava flows in the Western Ghats. However, the main part of the Mandla sequence and most likely significant parts of the Malwa series erupted synchronous with the Western Ghats series. Similar to the southwestern sequences, the Northern Province segments show their main activity during the magnetochron C29r. This sets a time span of 0.7 Ma for the main activity of the most parts of the Deccan LIP.
- As the Northern Province segments are spatially independent, they should not be correlated with younger but geochemically similar formations of the Western Ghats (e.g. Bushe, Poladpur or Ambenali Fm). The Northern Province shows clear differences in stratigraphy. Therefore it is necessary to establish a new subgroup for the Northern Deccan Province. We suggest to extend the term "Narmada Fm" introduced by Sreenivasa Rao et al. (1985) to the "Narmada Subgroup". A clear petrologic and geochemical characterization of the individual formations of the "Narmada Subgroup" needs then to be the focus of further investigations and is also in progress.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jseaes.2014.03.022.

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